Hospital Uptime

Tiered solar, battery and generator plans that keep a hospital's life-critical loads on through 8 to 20 hours of daily grid outage.

hospital-uptime.designmaster.workers.dev · github.com/DesignMaster-Solutions/hospital-uptime (MIT)

The problem

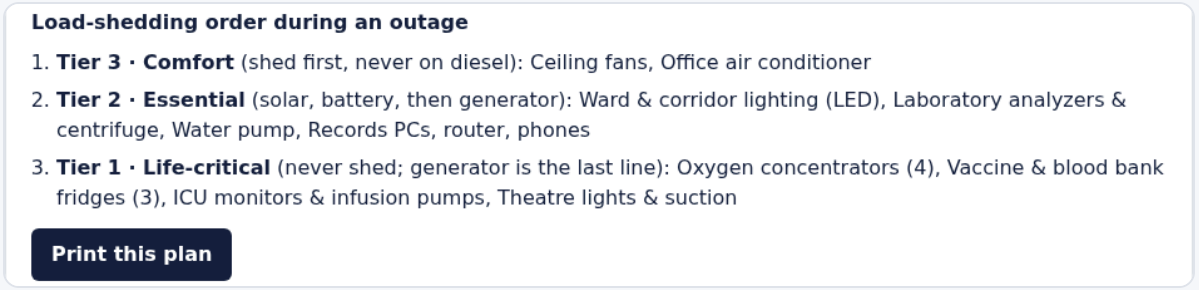
- Nigerian hospitals lose grid power **8 to 20 hours a day**.
- The default answer is one big diesel genset carrying **every** load, every outage hour.
- That costs fuel every day, depends on fuel supply, and when it fails, oxygen, vaccine fridges and ICU monitors fail with it.
- Administrators have no simple way to size a safer system or defend it to a lender.

The brief

"Design a scalable, low-maintenance energy system that guarantees uninterrupted 24-hour power for Nigerian hospitals experiencing 8–20 hours of daily grid outages."

The idea: power by tier, not by building

- **Tier 1, life-critical**: oxygen, vaccine and blood fridges, ICU, theatre. Never shed; the generator is the last line.
- **Tier 2, essential**: lighting, lab, water pump, records.
- **Tier 3, comfort**: fans, AC. Shed first, never on diesel.
- Solar carries the day, the battery carries the night, a **small** generator only backs tiers 1-2.



The product

1. List the equipment and its tier.
2. Set the grid: outage hours, start time, sun, diesel price.
3. Auto-size solar, battery and generator.
4. See every hour: which source carries each tier, what is shed.
5. Stress-test 8-20 h outages, sunny and cloudy.
6. See the money and print the plan.



Results for the sample hospital (14 h outage)

Life-critical energy unserved

0 kWh

Diesel

21 → 2.5 L/day

Simple payback

4.3 years

Stress test of the same system: tiers 1-2 stay on in all 8 cases (8, 12, 16, 20 h outage × normal / cloudy day). Diesel 0-4.4 L/day; only comfort load is shed (up to 12 kWh on a 20 h cloudy day).

Stress test: the same system across 8-20 h outages

Outage per day	Normal sun	Cloudy day
8 h	tiers 1-2 on 0 L diesel · 0 kWh comfort shed	tiers 1-2 on 0 L diesel · 0 kWh comfort shed
12 h	tiers 1-2 on 1.3 L diesel · 2.4 kWh comfort shed	tiers 1-2 on 1.3 L diesel · 2.4 kWh comfort shed
16 h	tiers 1-2 on 2.5 L diesel · 5.5 kWh comfort shed	tiers 1-2 on 3.9 L diesel · 7.2 kWh comfort shed
20 h	tiers 1-2 on 2.5 L diesel · 5.5 kWh comfort shed	tiers 1-2 on 4.4 L diesel · 12 kWh comfort shed

Cloudy = half the peak sun hours, every day of the outage block. Unserved energy per day; diesel only ever covers tiers 1 and 2.

How it works and how it is checked

- Load per line = W × h / 24, grouped by tier.
- Outage = one continuous block per day (worst case for storage).
- Solar = half-sine 06:00-18:00 × 0.75 performance ratio.
- Dispatch per tier: solar → battery → generator (tiers 1-2 only).
- 3 simulated days, the 3rd reported.
- 7 automated tests: auto-sized system never drops a life-critical load; comfort is shed first and never gets diesel; hybrid always burns less diesel than genset-only; stress range covered.
- Prices are editable planning assumptions, not quotes. Surge, cabling and protection stay with a licensed installer.

Path to impact, 6 to 18 months

- **Pilot** with one hospital: import real meter data instead of a typed equipment list.
- **Installer quotes** plugged into the cost model; one plan, comparable bids.
- **Lender-ready export** (monthly payment vs. diesel saved) for PAYG and asset finance.
- **Adapts anywhere**: outage pattern, sun, tariffs and prices are inputs, so the same tool serves clinics in Africa, the Americas and Asia.
- Free for hospitals; revenue from installers and lenders who receive qualified, pre-sized projects.

Team and disclosure



Alan De Bortolo, founder, Design Master Solutions LLC (solo entry)

Software for clinics and small businesses since 2004.

Repository started Oct 29, 2026, 22:06 EDT; everything in it was built for this hackathon. AI: Claude Code as coding assistant. No third-party code or datasets; sample hospital and prices are illustrative assumptions.